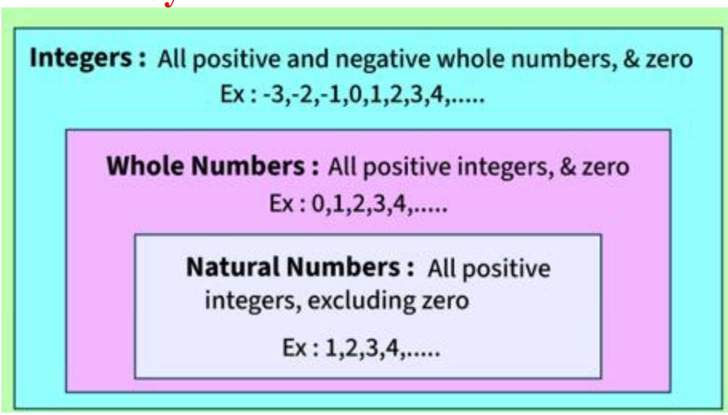
SURYA THE GLOBAL SCHOOL
CONCEPT REINFORCEMENT MATERIAL

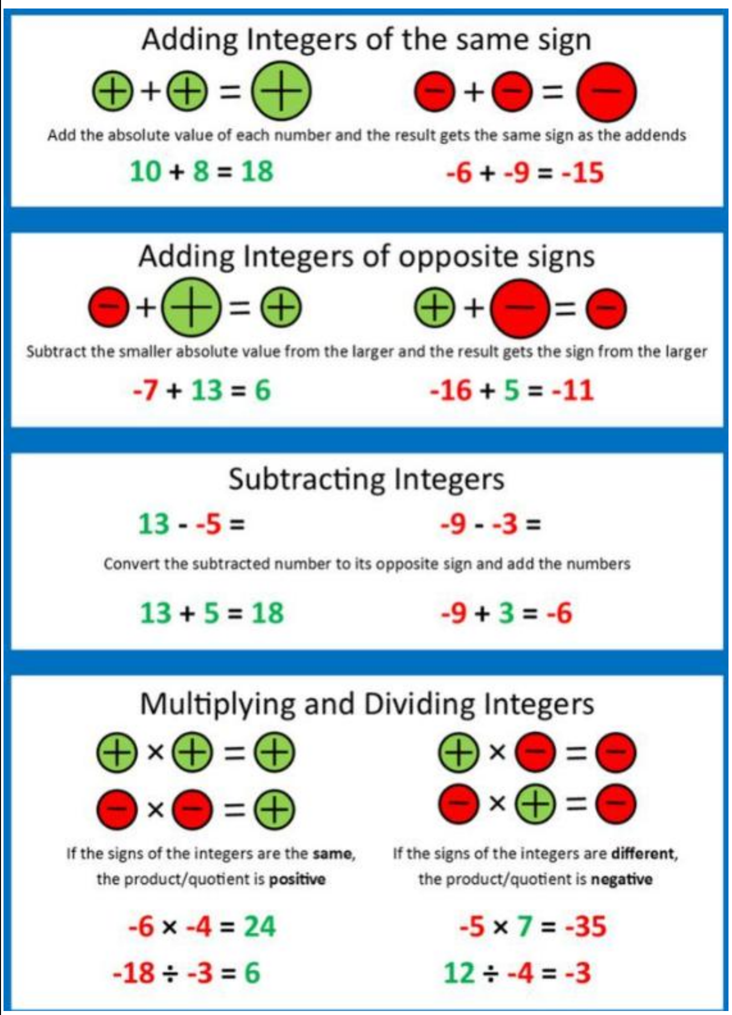
SUBJECT: MATHEMATICS
GRADE: VII

DAY-1(29-9-25)

Number System:



Operations on Integers:



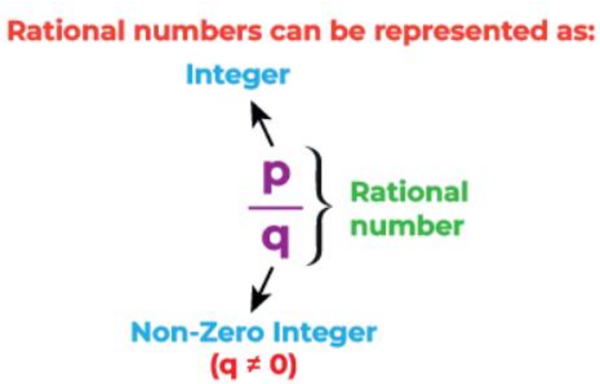
DAY-2(30-9-25)

Properties of Integers:

Property	Definition	Example
Closure	Sum, difference, or product of any 2 integers is an integer (not always for division)	$-4+6=2; 3\times-2=-6$
Commutative	Order doesn't change answer (addition, multiplication)	$5+8=8+5;$ $-3\times7=7\times-3$
Associative	Grouping doesn't change answer (addition, multiplication)	$(1+2)+3=1+(2+3)$
Distributive	Multiply over addition/subtraction	$2\times(3+4)=2\times3+2\times4$
Identity	Special number keeps integer same: 0 for addition, 1 for multiplication	$a+0=a; a\times1=a$

DAY-3(5-10-25)

Rational numbers:



Types of Rational numbers:

- **Positive Rational Numbers** → both numerator & denominator same sign (e.g., $\frac{5}{7}$, $\frac{-9}{-11}$)
- **Negative Rational Numbers** → numerator & denominator have opposite signs (e.g., $\frac{-3}{8}$, $\frac{7}{-12}$)

DAY-4(6-10-25)

Operations on Rational Numbers:

Addition:

$$\frac{2}{3} + \frac{5}{4}$$

Step 1: Since the denominator are not same, we need to take LCM.

LCM (4, 3) = 12

Step 2: Convert to the same denominators.

$$\frac{2}{3} + \frac{5}{4} = \left(\frac{2}{3}\right) \times \left(\frac{4}{4}\right) + \left(\frac{5}{4}\right) \times \left(\frac{4}{4}\right)$$

Add the numerators.

$$= \frac{8}{12} + \frac{15}{12}$$

Keep the denominator same.

$$= \frac{23}{12}$$

Subtraction:

Subtract the numerators.

$$\frac{7}{3} - \frac{5}{3} = \frac{7-5}{3} = \frac{2}{3}$$

Keep the denominator same.

$$\frac{2}{3} - \frac{1}{4}$$

Step 1: Since the denominator are not same, we need to take LCM.

LCM (3, 4) = 12

Step 2: Convert to the same denominators.

$$\frac{2}{3} - \frac{1}{4} = \left(\frac{2}{3}\right) \times \left(\frac{4}{4}\right) - \left(\frac{1}{4}\right) \times \left(\frac{3}{3}\right)$$

Subtract the numerators.

$$= \left(\frac{8}{12}\right) - \left(\frac{3}{12}\right)$$

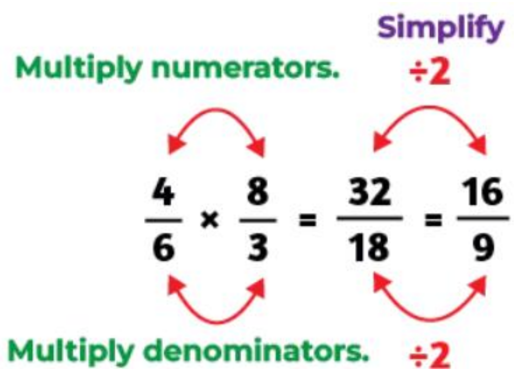
Keep the denominator same.

$$= \frac{(8-3)}{12}$$

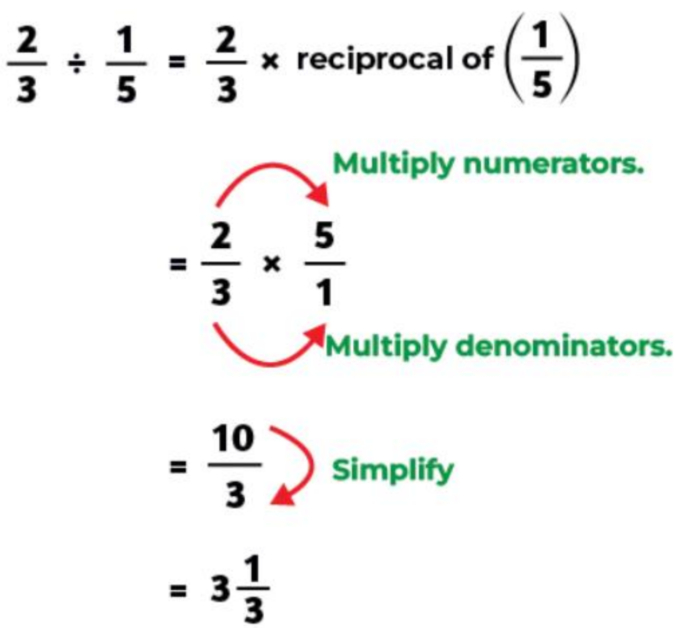
$$= \frac{5}{12}$$

DAY-5(7-10-25)

Multiplication:



Division:



DAY-6(8-10-25)

Factors and Multiples:

- **Factor**: A number that divides another number exactly.
  Example: Factors of 12 → 1, 2, 3, 4, 6, 12.
- **Multiple**: The product of a number and any whole number.
  Example: Multiples of 4 → 4, 8, 12, 16...
- **Prime Number**: A number with only two factors: 1 and itself.
  Example: 2, 3, 5, 7.
- **Composite Number**: A number with more than two factors.
  Example: 4, 6, 8, 9.

DAY-7(9-10-25)

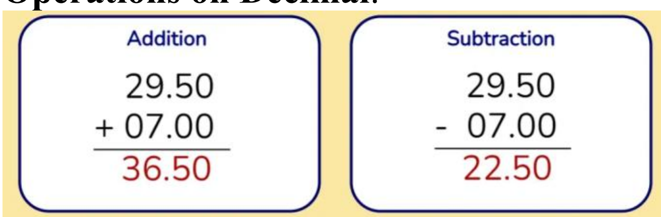
Decimals:

A decimal number is used to represent fractions and parts of a whole using a decimal point .

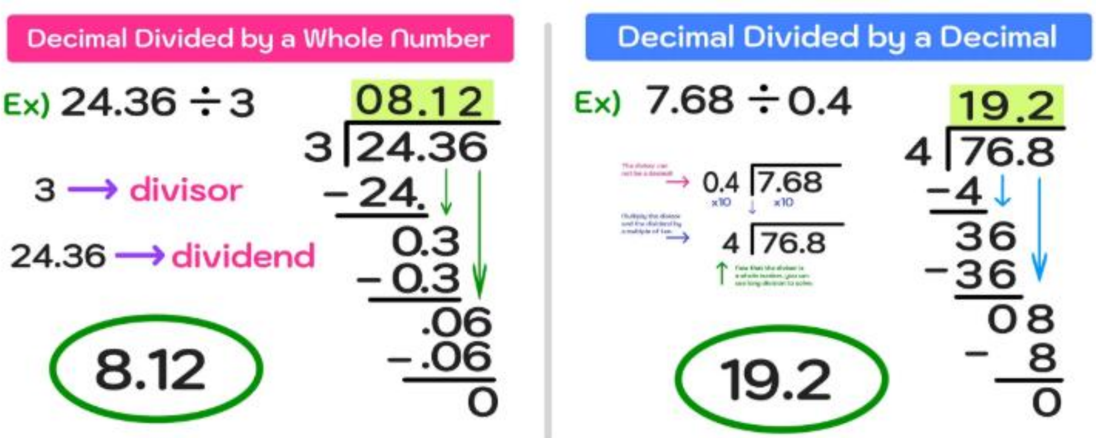
Examples: 0.5,2.75,13.0080.5, 2.75, 13.0080.5,2.75,13.008

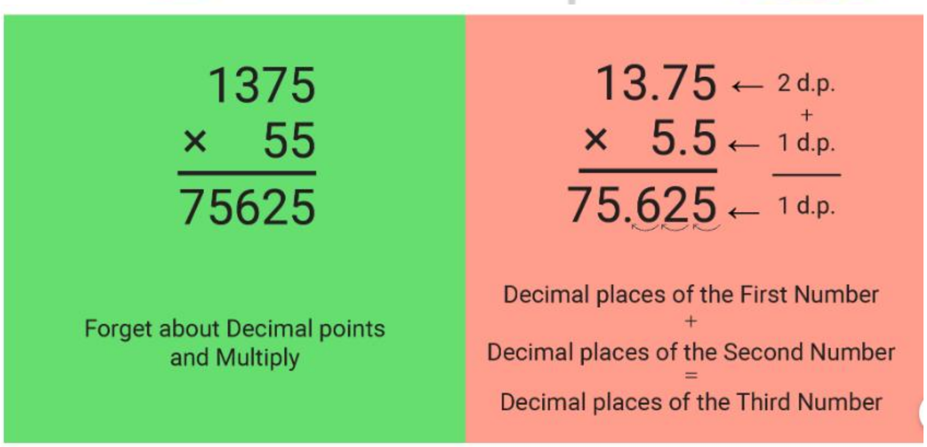
- Whole number part → left of the decimal point
- Decimal (fractional) part → right of the decimal point

Operations on Decimal:



DAY-8(10-10-25)



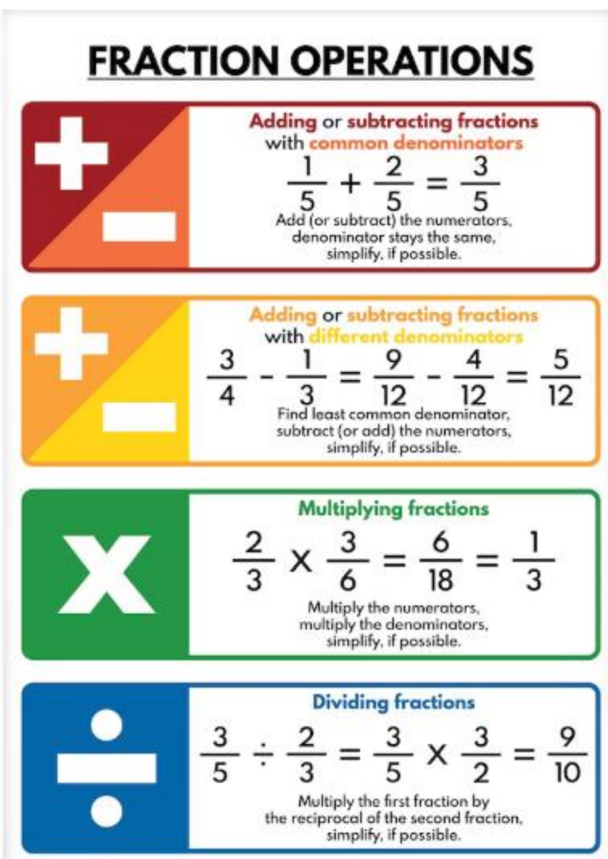


DAY-9(11-10-25)

Fractions: **Definition:** A number that represents a part of a whole.
Example: $\frac{3}{4}$ means 3 parts out of 4.

Types of fractions	Definition	Example
Unit fractions	Fractions with numerator 1 .	$\frac{1}{7}$
Proper Fractions	Fractions in which the numerator is less than the denominator.	$\frac{2}{7}$
Improper Fractions	Fractions in which the numerator is more than or equal to the denominator.	$\frac{5}{3}$
Mixed Fractions	Mixed fractions consist of a whole number along with a proper fraction.	$8\frac{2}{3}$
Like Fractions	Fractions with the same denominators.	$\frac{1}{4}$ and $\frac{3}{4}$
Unlike Fractions	Fractions with different denominators.	$\frac{1}{3}$ and $\frac{3}{4}$
Equivalent Fractions	Fractions that have the same value after being simplified or reduced.	$\frac{6}{4}$ and $\frac{12}{8}$

DAY-10(12-10-25)



DAY-11(13-10-25)

Measurement:(conversion of units)

Metric Units of Measure		
Length	**Capacity**	**Weight (mass)**
1 **kilo**meter (km) = 1000 meters (m)	1 **kilo**liter (kl) = 1000 liters (L)	1 **kilo**gram (kg) = 1000 grams(g)
1 **hecto**meter (hm) = 100 m	1 **hecto**liter (h1) = 100 m	1 **hecto**gram (hg) = 100 g
1 **deka**meter (dam) = 10 m	1 **deka**liter (dal) = 10 m	1 **deka**gram (dag) = 10 g
1 meter (m) = 1 m	**1 liter (L) = 1 L**	**1 gram (g) = 1 g**
1 **deci**meter (dm) = 0.1 m	1 **deci**liter (dl) = 0.1 L	1 **deci**gram (dg) = 0.1 g
1 **centi**meter (cm) = 0.01 m	1 **centi**liter (cl) = 0.01 L	1 **centi**gram (cg) = 0.01 g
1 **milli**meter (mm) = 0.001 m	1 **milli**liter (ml) = 0.001 L	1 **milli**gram (mg) = 0.001 g

DAY-12(14-10-25)

Ratio :

Ratio: A ratio is a way to compare two quantities of the same kind by division.

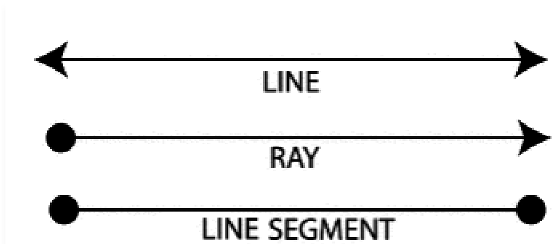
- Written as **a : b** (read as "a is to b").
- **Example:** In a basket, there are **8 apples** and **4 oranges**.
- Ratio of apples to oranges = 8:4=2:1
- This means → there are 2 apples for every 1 orange

Geometry :

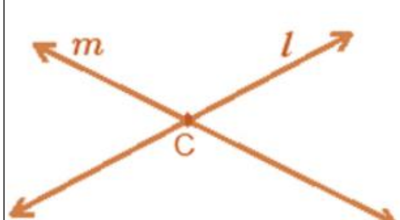
Line: If we take a point and draw a straight path that extends endlessly on both the sides, then the straight path is called as a line.

Ray A ray is a part of a line with one endpoint.

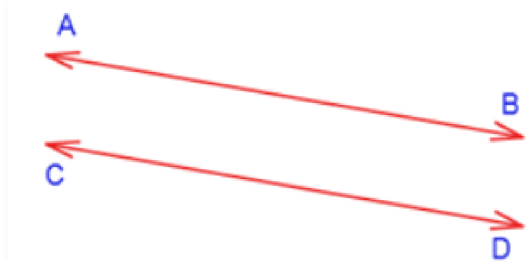
Line segment: A line segment is a part of a line with two endpoints.



Intersecting Lines: If two lines have a common point, then they are said to be intersecting.

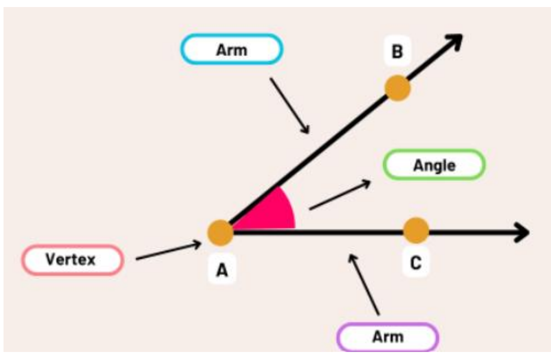


Parallel lines: If two lines have no common point, then they are said to be parallel. Here AB and CD are parallel lines.

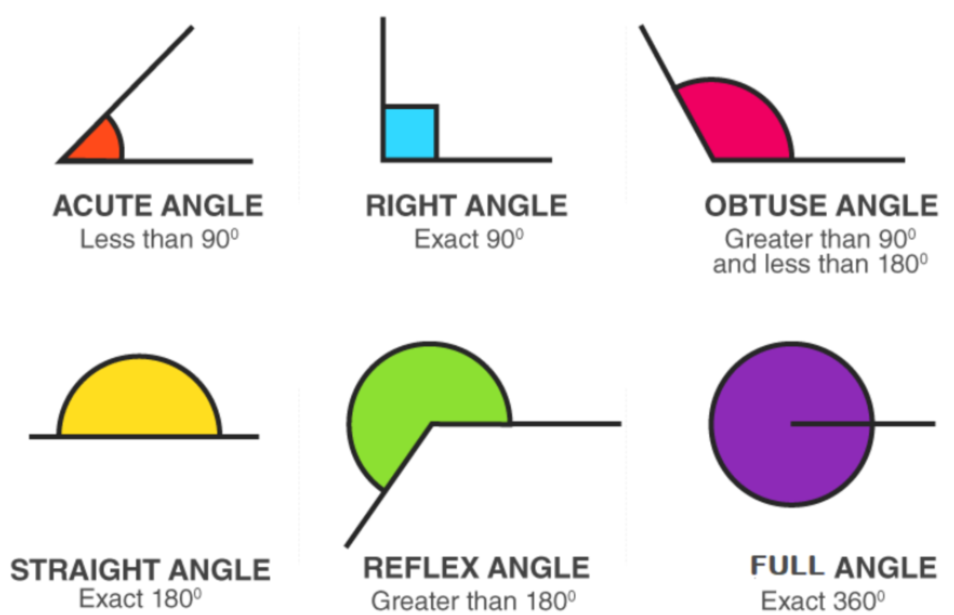


DAY-13(15-10-25)

Angle:

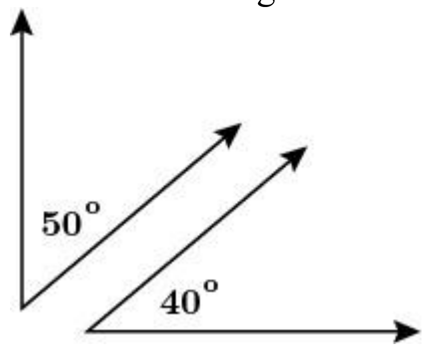


- **Types of Angles:**



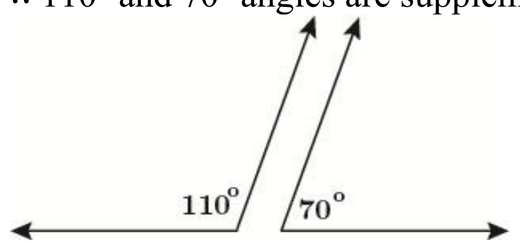
Complementary Angles:

Two angles whose sum is 90^0 are called complementary angles.
Example: $50^0 + 40^0 = 90^0$
$\therefore$ 50^0 and 40^0 angles are complementary angles.



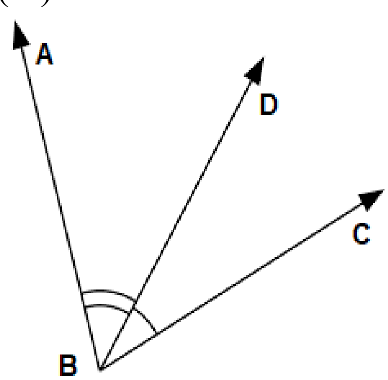
Supplementary angles

- Two angles whose sum is 180^0 are called supplementary angles.
  Example: $110^0+70^0=90^0$
  $\therefore$ 110^0 and 70^0 angles are supplementary angles.



Adjacent Angles:

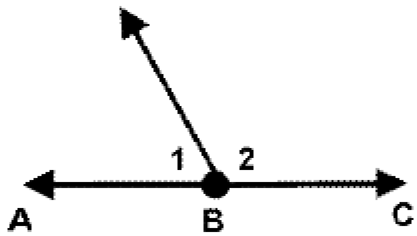
- Two angles are adjacent, if they have
  (i) A common vertex
  (ii) A common arm
  (iii) Their non-common arms on different sides of the common arm.



- Here ∠ABD and ∠DBC are adjacent angles.

Linear Pair:

- Linear pair of angles are adjacent angles whose sum is equal to 180∘.

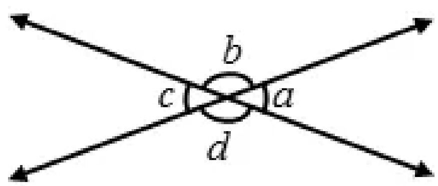


Here, 1 and 2 are linear pair of angles.

DAY-14(16-10-25)

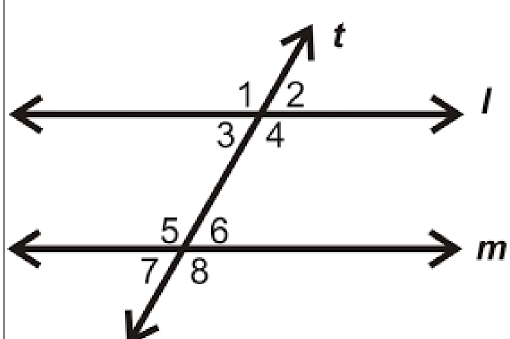
Vertically Opposite Angles:

- Vertically opposite angles are formed when two straight lines intersect each other at a common point.
- Vertically opposite angles are equal.



Here, the following pairs of angles are vertically opposite angles.
(i) a and c
(ii) b and d

Transversal of Parallel Lines:



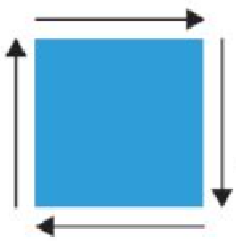
- If a transversal intersects two parallel lines, then each pair of corresponding angles is equal.
  (i) ∠1=∠5 (ii) ∠2=∠6
  (iii) ∠3=∠7 (iv) ∠4=∠8

- If a transversal intersects two parallel lines, then each pair of alternate interior angles is equal.
  (i) $\angle 3=\angle 6$ (ii) $\angle 4=\angle 5$
- If a transversal intersects two parallel lines, then each pair of interior angles on the same side of the transversal is supplementary.
  (i) $\angle 3+\angle 5=180^0$ (ii) $\angle 4+\angle 6=180^0$

DAY-15(17-10-25)

Perimeter & Area:

- **Perimeter**: Distance around a shape.



Perimeter of a rectangle = 2 x (length + breadth)

Example:

Length = 12 cm, Breadth = 8 cm

$$\text{Perimeter} = 2 \times (12 + 8) = 2 \times 20 = 40\,\text{cm}$$

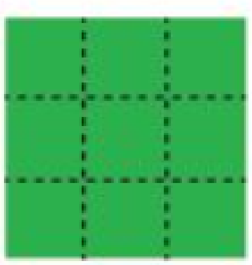
Perimeter of a square =4×side

Example:

Side = 7 cm

$$\text{Perimeter} = 4 \times 7 = 28\,\text{cm}$$

Area: Space inside a shape.



Area of a square = side × side.

Example:

Side = 6 cm

$$\text{Area} = 6 \times 6 = 36\,\text{cm}^2$$

Area of Rectangle=Length x Breadth

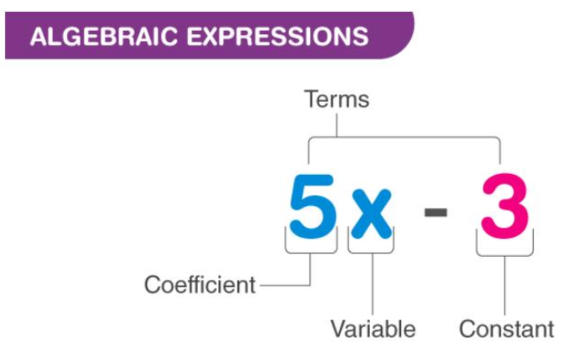
Example:

Length = 12 cm, Breadth = 5 cm

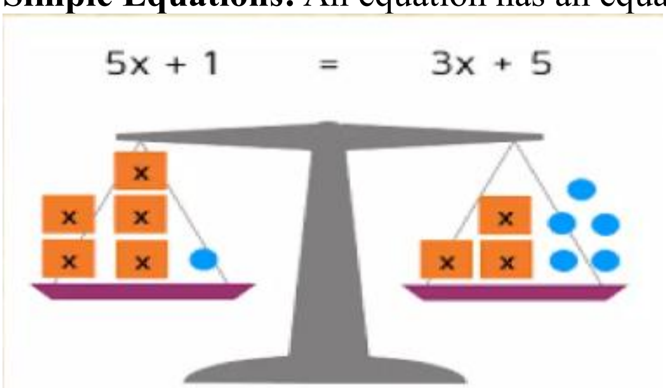
$$\text{Area} = 12 \times 5 = 60\,\text{cm}^2$$

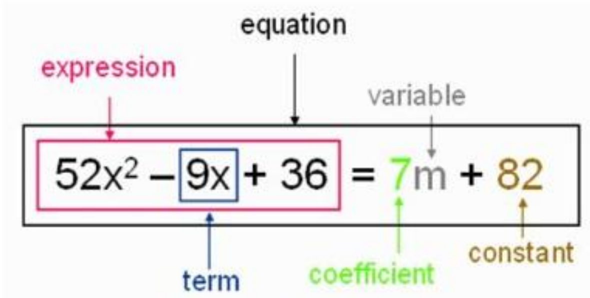
DAY-16(18-10-25)

Algebra :



Simple Equations: An equation has an equal sign =.

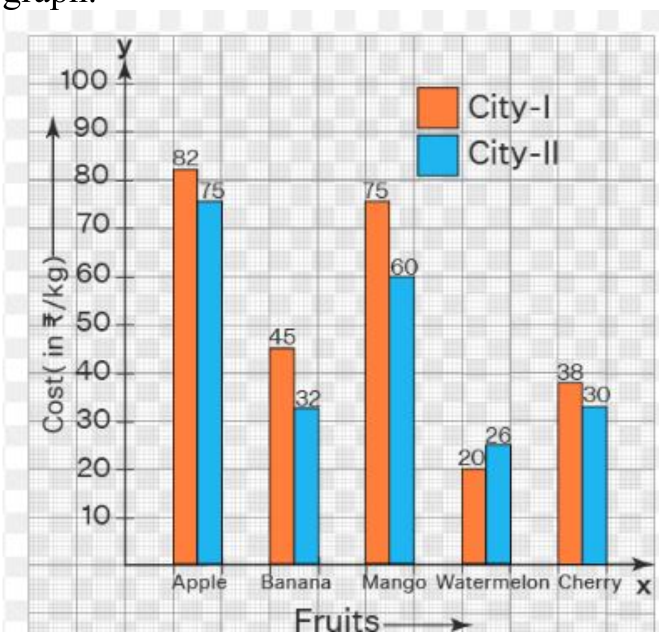




DAY-17(19-10-25)

Data Handling:

Double bar graph: **A double bar graph** is used to display two sets of data on the same graph.



Mean (Average): The mean is the sum of all values divided by the number of values.

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Example:

The marks obtained by 5 students in a test are: 12, 18, 20, 15, and 25.
Find the mean of their marks.

Solution:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

$$\text{Sum of marks} = 12 + 18 + 20 + 15 + 25 = 90$$

$$\text{Number of students} = 5$$

$$\text{Mean} = \frac{90}{5} = 18$$

Median (Middle Value): The **median** is the middle value when the data is arranged in order.

- If the number of values is **odd** → middle value.
- If the number of values is **even** → average of the two middle values.

Example:1 Find the median of the numbers: 7, 3, 9, 5, 11

Solution:
Step 1: Arrange in ascending order → 3, 5, 7, 9, 11
Step 2: Number of observations = 5 (odd)

Middle value = 3rd number = **7**

Answer: Median = **7**

Example:2 Find the median of the numbers:
12, 6, 10, 4

Solution:
Step 1: Arrange in ascending order → 4, 6, 10, 12
Step 2: Number of observations = 4 (even)

$$\text{Median} = \frac{\text{Middle two terms}}{2}$$

Middle terms = 6 and 10

$$\text{Median} = \frac{6 + 10}{2} = \frac{16}{2} = 8$$

Mode (Most Frequent Value): The **mode** is the value that occurs most often.

Example 1 (Single Mode):

Find the mode of:
4, 7, 9, 4, 3, 4, 5, 7

Solution:

- Number 4 occurs **3 times**
- Number 7 occurs **2 times**
- Other numbers occur once

So, **Mode = 4**

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